STAFF REVIEW · H446 106 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. RAM is local working memory; virtual storage presents remote/pooled file storage.

2. In its left subtree (under the chosen no-duplicate convention).

3. Its element locations can be calculated from base address and index.

4. Sum 0 and carry-out 1.

I Do — source worked steps

1. Model answer — The OS allocates CPU time to both processes; the file system writes the document to storage; a device driver controls sound output. Memory management keeps process data protected. — Check: link each claim to the scenario and use precise OCR terminology.

We Do — guided CHECK

1. What must we decide?
Identify the inputs, relevant facts and exact output or judgement required.
Work through the method
Use the middle stage of the visual to follow the process and state each intermediate result.
Shared worked answer
Identify concurrent processes; explain the print queue/device driver; describe file and network access. Printing and downloading share resources without either application controlling the hardware directly.

You Do — source mark schemes

1. Up to 2 marks for each explained service: process scheduling, memory management, file management, device management, security or interface.

2. It manages a print queue/spooling [2], coordinates each print job in turn [2], and uses a driver to communicate with the printer [2].

3. Identify need for abstraction and drivers [2], arbitration of shared hardware [2], memory/process protection [2], and a reasoned conclusion with example [2].

Exit Ticket — source answer

1. False. It also manages processes, memory, files, devices and security.